

4 States of matter

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Ideal gas particles occupy negligible volume and have no intermolecular attraction. $pV = nRT$ uses SI units and kelvin. Real gases deviate most at high pressure or low temperature where particle volume and attraction matter. Solids, liquids and gases differ in particle arrangement, movement and proximity, while intermolecular forces influence melting and boiling.

Worked example: follow the reasoning

0.10 mol gas at 300 K occupies $2.49 \times 10^{-3} \text{ m}^3$ at 100 kPa, since $V = nRT/p$.

Quick check · multiple choice

A Which temperature must be used in $pV=nRT$?

[1]

- A Degrees Celsius
- B Degrees Fahrenheit
- C Any unconverted scale
- D Kelvin

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Calculate pressure of 0.050 mol ideal gas at 300 K in 1.0 dm^3 ($R = 8.31$).

[3]

Final answer: _____

Answers and explanation

Quick check: D. Kelvin

Question 1 [3 marks]

$V = 0.0010 \text{ m}^3$ [1]; $p = nRT/V$ [1] = $1.25 \times 10^5 \text{ Pa}$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.